6,906: Economics: Macro for Computer Science

Lecture 6: Goods and financial markets

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Learning objectives

- By the end of this lecture, you should be able to:
  - Describe how demand determines output in the short run and the factors that affect demand;
  - Explain how, in the short run, the central bank determines the interest rate.

The study of economic activity in the short run

- We aim to understand how output and the interest rate are determined in the short run.
- To this end, we look at the interaction between the financial market and the market for goods and services.
- John Hicks and Alvin Hansen called this framework the IS-LM model, which is a mathematical representation of the ideas expressed by John Maynard Keynes in the book "General Theory of Employment, Interest and Money."
- We begin with the **basic IS-LM model**. The steps we follow are to describe:
  - This lecture
    - The goods and services market and the IS relation
    - Financial markets and the LM relation
  - Next lecture
    - The intersection of the IS and the LM relations
    - Policy options in the IS-LM model

The demand for goods and services

- In the short run, prices are rigid (do not change). For this reason, in the short run, demand determines output as production adjusts automatically to demand without price changes.
- The mechanism is the following:
  - Changes in the demand for goods and services lead to changes in production;
  - Changes in production lead to changes in income;
  - Changes in income lead to changes in the demand for goods.

The demand for goods and services

- Starting from the expenditure approach to compute GDP, the total demand for goods and services Z is defined as

$$Z = C + I + G + X - IM.$$

- Under the assumption of a closed economy, $X = IM = 0$, total demand is

$$Z = C + I + G.$$

Components of GDP - Private consumption of goods and services

- There are two prominent theories of how consumption is determined
  - The Keynesian theory states that consumption depends only on current disposable income. The main critique is that it is postulated from aggregated data and not micro-funded. It doesn't consider important factors such as income shifts over the life cycle, future income or savings, wealth, etc.
  - The permanent income hypothesis (PIH) theory states that consumption depends on permanent income (i.e., expected long-term average income). It assumes that households choose their consumption path to maximize utility, considering their intertemporal budget constraints.
- In this course, we take the Keynesian approach.

Components of GDP - Private consumption of goods and services

- Consumption is defined as function of disposable income $C(Y_D)$, where disposable income is $Y_D = Y - T$, with Y defining income and T taxes or government transfers.

$$C = C(Y_D).$$
$$(+)$$

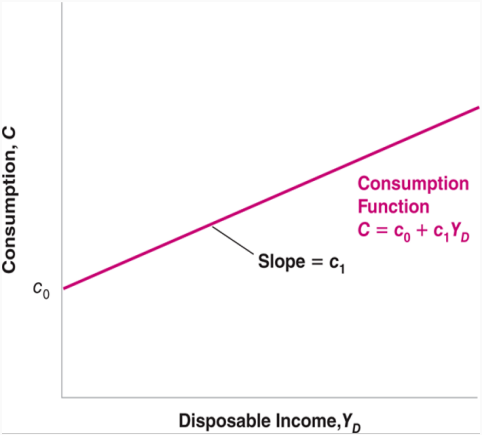
- We model the consumption function to be a linear function

$$C = c_0 + c_1 Y_D = c_0 + c_1(Y - T),$$

with two parameters - the autonomous consumption c_0 and the (marginal) propensity to consume $0 < c_1 < 1$.

Components of GDP - Private consumption of goods and services

- The autonomous consumption c_0 is what people would consume if their disposable income equals zero. .
- The propensity to consume c_1 gives the effect an additional unit of disposable income has on consumption.



Source: Blanchard (2021)

Components of GDP - Private investment

- Private investment comprises investment in equipment and construction.
- It is an important component of GDP as it determines its level and short-run fluctuations, being the most volatile component of GDP.
- Because it contributes to capital accumulation, it's also important for long-term growth, as seen in Part II.
- Investment depends mainly on two factors: production Y (or sales) and the interest rate i

$$I = I(Y, i).$$
$$(+, -)$$

- For now, we assume investment is fixed at level $\bar{I}$, which makes it an exogenous variable since it does not depend on other variables.
- Remember that endogenous variables are explained within the model, while exogenous variables are taken as given.

Components of GDP - Government purchases of goods and services

- The government's choice of taxes and/or transfer payments (T) and spending on goods and services G represent the fiscal policy.
- In this course, following Blanchard's book, we treat G and T as exogenous because:
  - Fiscal policy changes do not happen with the same regularity as consumers' or firms' actions.
  - G and T are variables chosen by the government and not explained within the model.

Equilibrium in the goods and services

- In equilibrium, production Y equals demand Z

$$Y = Z \Leftrightarrow Y = C(Y_D) + \bar{I} + G = c_0 + c_1(Y - T) + \bar{I} + G.$$

- In equilibrium, production equals demand, which in turn depends on income, which is itself equal to production.
- An alternative way of expressing this equilibrium condition is that saving, which is the sum of private saving $Y - T - C$ and public saving $T - G$, equals investment.
- Stating the equilibrium condition this way explains its name as the **investment - savings (IS) relation.**

Equilibrium in the goods and services

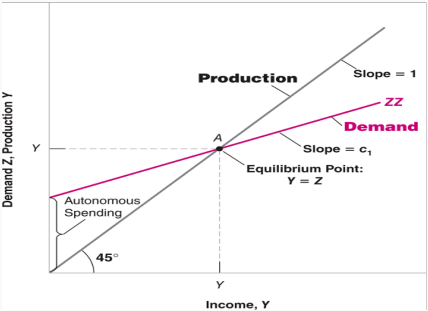
- The equilibrium conditions can be rewritten as:

$$Y = \frac{1}{1 - c_1}(c_0 - c_1 T + \bar{I} + G),$$

  where $(c_0 - c_1 T + \bar{I} + G)$ is defined as autonomous spending and $\frac{1}{1-c_1}$ as the multiplier since it is larger than 1.
- Autonomous spending is positive because if $T = G$ (balanced budget) and c_1 is between 0 and 1, then $(G - c_1 T)$ is positive, and so is autonomous spending.
- The multiplier is larger when c_1 is closer to 1.

Equilibrium in the goods and services

- Production as a function of income coincides with the 45-degree line because production equals income.
- The demand for goods and services, Z, is an increasing function of output.
- ZZ is flatter than the 45-degree line because we assume that an increase in output leads to a less than one-for-one increase in demand as empirical evidence suggests.
- The intersection of ZZ and the 45-degree line is the equilibrium level of output.



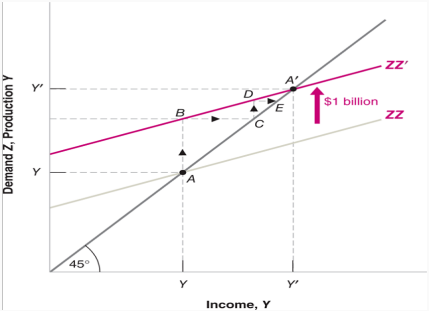
Source: Blanchard (2021)

Equilibrium in the goods and services

- The demand for goods and services $Z = c_0 + c_1(Y - T) + \bar{I} + G$ increases if
  - autonomous consumption c_0 is increased
  - government spending G is increased
  - taxes T are reduced

Equilibrium in the goods and services

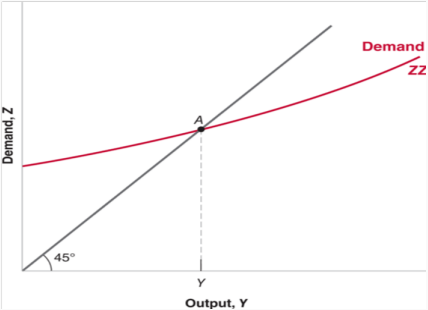
- How does the equilibrium change if autonomous consumption increases?
- An increase in demand leads to an increase in production and income, which in turn leads to a future increase in demand...
- The increase after $n+1$ rounds is $1 + c_1 + c_1^2 + c_1^3 + ... + c_1^n$, which is a geometric series with a limit of $\frac{1}{1-c_1}$.
- If c_1 equals 0.5, the multiplier equals 2, meaning that an increase in autonomous consumption by CHF 1 billion will increase output by CHF 2 billion.



Source: Blanchard (2021)

Equilibrium in the goods and services

- The plot for the general demand function, $Z = C(Y - T) + I(Y, i) + G$. is slightly different.
- ZZ is a curve rather than a line because we have not assumed that the investment relation is linear.
- ZZ is upward-sloping because, for a given value of the interest rate, an increase in output leads to an increase in the demand for goods and services through its effects on consumption and investment.



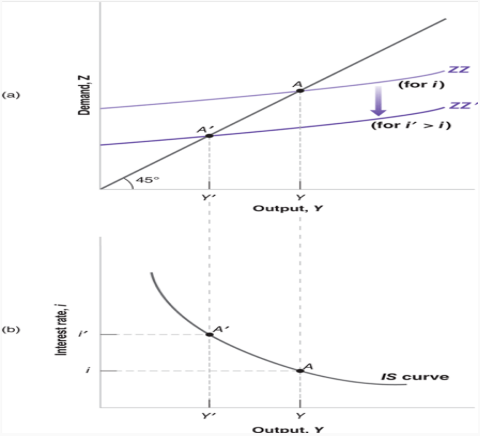
Source: Blanchard (2021)

The derivation of the IS curve

$$Z = C(Y - T) + I(Y, i) + G.$$

(a) An increase in the interest rate decreases the demand for goods and services at any level of output, leading to a decrease in the equilibrium level of output.

(b) Equilibrium in the goods and services implies that an increase in the interest rate leads to a decrease in output. The IS curve is therefore downward sloping.



Source: Blanchard (2021)

Financial markets

- We start from the assumption that the financial markets include only two assets: money and bonds.
- Money is used for transactions, but it pays no interest.
- Bonds pay a positive interest rate, i (the rate of interest), but cannot be used for transactions.
- At every point in time, households choose how to allocate their financial wealth between money and bonds.

Money demand

- We describe the money market with supply and demand.
- Demand for money is equal to nominal income $\$Y$ (a measure of the level of transactions in the economy) times a decreasing function of the interest rate i: $M^d = \$YL(i)$. Divide both sides of the equation by the price level P:

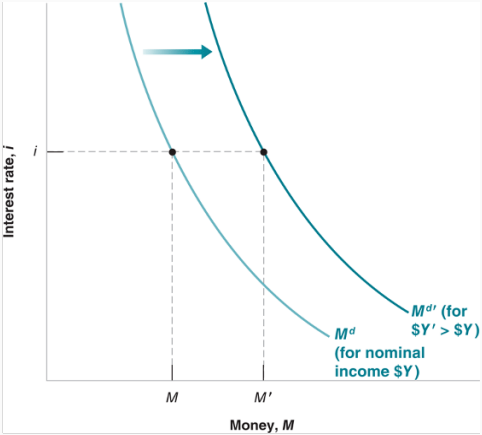
$$\frac{M^d}{P} = YL(i),$$

$$(+, -)$$

- where M^d is money demand, Y income, and $L(i)$ liquidity demand, which is a function of interest rate i.

Money demand

- If the interest rate goes up you put more into bonds (money bears no interest);
- If your income goes up, you consume more now. To do this, you need more money.



Source: Blanchard (2021)

Money supply and equilibrium in financial markets

- Suppose the central bank decides to supply an amount of money equal to M:

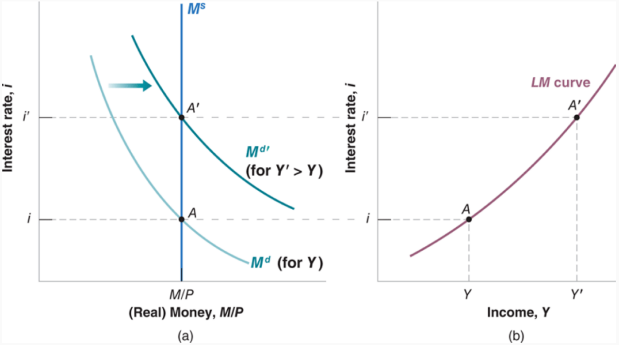
$$M^s = M$$

- Equilibrium in financial markets requires that $M^s = M^d = M$.
- This implies that, in equilibrium, real money supply $\frac{M^s}{P}$ equals the real money demand $\frac{M^d}{P}$, which depends on real income Y, and the interest rate i:

$$\frac{M^s}{P} = \frac{M^d}{P} = \frac{M}{P} = YL(i).$$

- This is the **liquidity preference - money supply (LM) relation.** It gives us the relation between the interest rate and income for a given money supply.

The derivation of the "traditional" LM curve

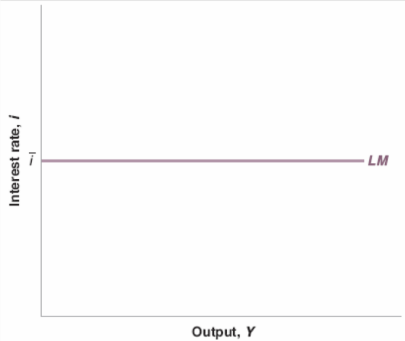


Source: Blanchard (2021)

- What happens if income increases? Suppose $Y' > Y$. This raises money demand. As M^s/P is fixed $M^s < M^d$ and L needs to fall. Thus, the interest rate increases, which implies that the LM curve slopes upwards in the (i,Y) space.

The horizontal LM curve

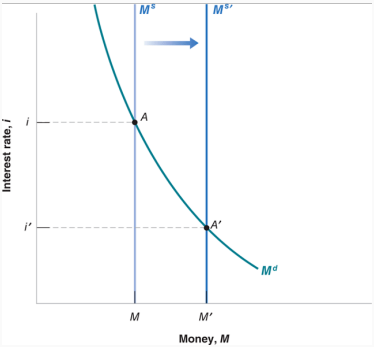
- The assumption that the central bank keeps the money stock constant and lets the interest rate adjust when income changes is not in line with what modern central banks do. Most central banks instead set the interest rate and adjust the money supply to achieve it.
- Hence, an increase in output will not lead to an increase in the interest rate unless the central bank chooses to raise its central bank rate. In the face of an increase in income, the central bank may increase its money supply and thus keep the interest rate constant.



Source: Blanchard (2021)

Determining the interest rate

- An increase in the supply of money leads to a decrease in the interest rate, hence in order to increase/decrease the interest rate, the central bank decreases/increases its supply of money.
- Central banks typically change the supply of money by buying or selling bonds in the bond market - open market operations.
- Expansionary open market operation: the central bank expands the supply of money by buying bonds.
- Contractionary open market operation: the central bank contracts the supply of money by selling bonds.



Source: Blanchard (2021)

Summary

- Today we learned:
  - To characterize the financial market and the market for goods and services of an economy.
  - To describe the equilibrium of the financial market and of the market for goods and services in the interest/income diagram.
- Next class, we will study the IS-LM model, which characterizes the implications of equilibrium in both the market for goods and services and the financial markets.